Taha Bekar

Robotics Software Engineer — Perception — SLAM — Autonomous Systems

mtahabekar@gmail.com — +1 (838) 262 0019

LinkedIn — Github

Education

Rensselaer Polytechnic Institute — Troy, NY

M.S. Electrical, Computer, and Systems Engineering — 2024 – May 2026

Focus: Autonomy, Robotics, and Control

Middle East Technical University — Ankara, Turkey

B.S. Electrical and Electronics Engineering — 2018 – 2024

GPA: 3.56 / 4.00

Research Experience

Deformable Object Manipulation Research – RPI — 2025 – 2026

Research Assistant / Master's Project

- Developed a ROS2-compatible deformable-object simulation framework in NVIDIA Isaac Sim using Newton Style3D for real-time cloth manipulation with anisotropic material, contact, friction, damping, and bending parameters.
- Built an obstacle-aware cloth manipulation pipeline with kinematic corner pinning, simulated cloth state publishing, ROS2 command interfaces, obstacle-distance computation, and closed-loop follower-corner control.
- Integrated a Control Barrier Function safety controller for cloth-obstacle avoidance for contact-rich manipulation scenarios.
- Developed real-time perception and state estimation pipeline for planar deformable objects using RGB-D camera data, ArUco marker tracking, and Kalman filtering in ROS for real-to-sim calibration.

Work Experience

Kuartis — Ankara, Turkey

Computer Vision / Machine Learning Engineer — Feb 2024 – Jun 2024

- Evaluated LiDAR point cloud registration algorithms including ICP, NDT, GICP, and FAST-GICP for SLAM pipelines and real-time mapping systems.
- Implemented LiDAR–IMU sensor fusion by combining LiDAR odometry with IMU odometry, evaluating both tightly coupled and loosely coupled fusion strategies for improved localization robustness.
- Analyzed performance of SLAM frameworks including LOAM, LIO-SAM, and FAST-LIO for localization accuracy and computational efficiency.
- Improved LIO-SAM localization accuracy by 6% by integrating GICP-based scan matching into the GTSAM factor-graph optimization framework.

GT-ARC Gemeinnützige GmbH, DAI Labor, TU Berlin — Berlin, Germany

Software Engineer Intern — Jul 2023 – Sep 2023

- Developed LiDAR-based free parking space detection algorithm for autonomous vehicles using ROS, Autoware, and HD Maps by implementing geometric intersection analysis between detected vehicles and parking regions.
- Achieved 86% free space detection accuracy.

Analog Devices — Istanbul, Turkey

Software Engineer Intern — Jul 2022 – Aug 2022

- Implemented PWM controller firmware on MAX32666 microcontroller using timer peripherals.
- Built GUI interface for testing PWM signals and implemented unit tests using Google Test.

Projects

Autonomous UGV Mapping System

- Converted a 2D LiDAR into a 3D scanning system via servo rotation to enable volumetric mapping.
- Implemented octomap mapping using ICP (Iterative Closest Point) registration and AMCL (Adaptive Monte Carlo Localization) algorithms in ROS. Achieved accuracy levels of 95% and 75% for mapping and localization in ROS.

Lane Detection System

- Implemented classical vision-based lane detection using Canny and Hough transform.

FPGA Network QoS Controller

- Implemented priority-based packet scheduling system in Verilog.

Technical Skills

Programming: Python, C++, C, MATLAB

Robotics: ROS, ROS 2, SLAM, NVIDIA Isaac Sim, MuJoCo, PhysX, Newton, Gazebo, Motion Planning, Localization, Kinematics

Perception: Computer Vision, LiDAR Point Cloud Processing

Libraries: PCL, OpenCV, PyTorch, NumPy

Tools: Linux, Git, Docker

Relevant Coursework

Safe Autonomy Computational Vision Machine Learning Robotics II Computational Optimization